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SQL> show user;

SQL> show con_name;

```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

show user;: Menampilkan user yang sedang terhubung

show con_name;: Menampilkan sesi ini aktif di PDB/ CDB

SQL> create tablespace ts1;

```
SQL> create tablespace ts1 datafile 'ts1

Tablespace created.
```

membuat tablespace baru bernama ts1

SQL> SELECT DBMS_METADATA.GET_DDL('TABLESPACE','TS1') FROM dual;

```
SQL> SELECT DBMS_METADATA.GET_DDL('TABLESPACE','TS1') FROM dual;

DBMS_METADATA.GET_DDL('TABLESPACE','TS1')
--------------------------------------------------------------------------------

  CREATE BIGFILE TABLESPACE "TS1" DATAFILE
  '/opt/oracle/product/23ai/dbhomeF
```

mengambil dan menampilkan kode DDL (Data Definition Language) atau skrip asli pembuatan tablespace 'TS1' untuk melihat secara lengkapnya

SQL> select * from dba_tablespaces

where tablespace_name='TS1' ;

```
DEF_INMEMORY_SERVICE_NAME
--------------------------------------------------------------------------------
LOST_WR C
------- -


TABLESPACE_NAME                BLOCK_SIZE INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS
------------------------------ ---------- -------------- ----------- -----------
MAX_EXTENTS   MAX_SIZE PCT_INCREASE MIN_EXTLEN STATUS    CONTENTS
----------- ---------- ------------ ---------- --------- ---------------------
LOGGING   FOR EXTENT_MAN ALLOCATIO PLU SEGMEN DEF_TAB_ RETENTION   BIG PREDICA
--------- --- ---------- --------- --- ------ -------- ----------- --- -------
ENC COMPRESS_FOR                   DEF_INME DEF_INME DEF_INMEMORY_DI
--- ------------------------------ -------- -------- ---------------
DEF_INMEMORY_COMP DEF_INMEMORY_ SHARED        DEF_INDE INDEX_COMPRES
----------------- ------------- ------------- -------- -------------
DEF_CELLMEMORY DEF_INMEMORY
-------------- ------------
DEF_INMEMORY_SERVICE_NAME
--------------------------------------------------------------------------------
LOST_WR C
------- -
OFF     N

TABLESPACE_NAME                BLOCK_SIZE INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS
------------------------------ ---------- -------------- ----------- -----------
MAX_EXTENTS   MAX_SIZE PCT_INCREASE MIN_EXTLEN STATUS    CONTENTS
----------- ---------- ------------ ---------- --------- ---------------------
LOGGING   FOR EXTENT_MAN ALLOCATIO PLU SEGMEN DEF_TAB_ RETENTION   BIG PREDICA
--------- --- ---------- --------- --- ------ -------- ----------- --- -------
ENC COMPRESS_FOR                   DEF_INME DEF_INME DEF_INMEMORY_DI
--- ------------------------------ -------- -------- ---------------
DEF_INMEMORY_COMP DEF_INMEMORY_ SHARED        DEF_INDE INDEX_COMPRES
----------------- ------------- ------------- -------- -------------
DEF_CELLMEMORY DEF_INMEMORY
-------------- ------------
DEF_INMEMORY_SERVICE_NAME
--------------------------------------------------------------------------------
LOST_WR C
------- -
```

menampilkan detail tablespace TS1 yang sudah dibuat

--REMEMBER THE RELATION

--TABLESPACE-SEGMENT-EXTENT-BLOCK

SQL> CREATE USER alex identified by alex

default tablespace ts1;

```
SQL> drop user ALEX;

User dropped.
```

```
SQL> CREATE USER alex identified by alex
default tablespace ts1;
  2
User created.
```

Membuat user baru bernama alex dengan password alex, serta menetapkan ts1 sebagai tempat penyimpanan data defaultnya

SQL> grant create session, create table, unlimited tablespace to alex;

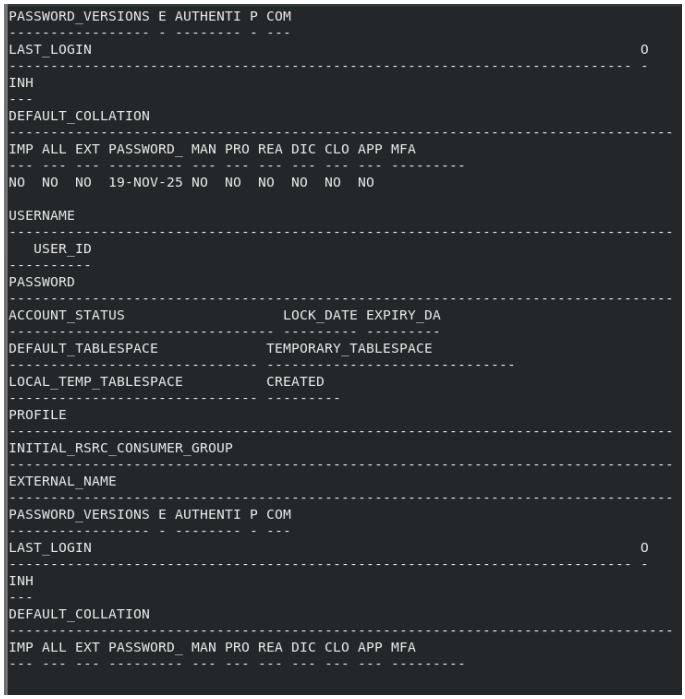
```
SQL> grant create session, create table, unlimited tablespace to alex;

Grant succeeded.
```

memberikan hak istimewa (grant) kepada user bernama alex untuk membuat sesi koneksi (login), membuat tabel baru, dan menggunakan ruang penyimpanan tablespace tanpa batas

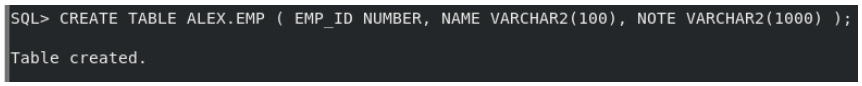
SQL> select * from dba_users

where username='ALEX';



mengambil dan menampilkan semua informasi mengenai user bernama ALEX yang diambil dari data dictionary view DBA_USERS

SQL> CREATE TABLE ALEX.EMP (EMP_ID NUMBER, NAME VARCHAR2(100), NOTE VARCHAR2(1000));



membuat sebuah tabel baru bernama EMP di dalam user ALEX. Tabel ini memiliki tiga kolom: EMP_ID (NUMBER), NAME (VARCHAR2(100)), dan NOTE (VARCHAR2(1000))

SQL> SELECT * FROM DBA_TABLES

WHERE TABLE_NAME='EMP'

AND OWNER='ALEX';

```
OWNER
--------------------------------------------------------------------------------
TABLE_NAME
--------------------------------------------------------------------------------
TABLESPACE_NAME
------------------------------
CLUSTER_NAME
--------------------------------------------------------------------------------
IOT_NAME
--------------------------------------------------------------------------------
STATUS     PCT_FREE   PCT_USED  INI_TRANS  MAX_TRANS INITIAL_EXTENT NEXT_EXTENT
-------- ---------- ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS PCT_INCREASE  FREELISTS FREELIST_GROUPS LOG B   NUM_ROWS
----------- ----------- ------------ ---------- --------------- --- - ----------
    BLOCKS EMPTY_BLOCKS  AVG_SPACE  CHAIN_CNT AVG_ROW_LEN
---------- ------------ ---------- ---------- -----------
AVG_SPACE_FREELIST_BLOCKS NUM_FREELIST_BLOCKS DEGREE     INSTANCES  CACHE
------------------------- ------------------- ---------- ---------- -----
TABLE_LO SAMPLE_SIZE LAST_ANAL PAR IOT_TYPE     T S NES BUFFER_ FLASH_C CELL_FL
-------- ----------- --------- --- ------------ - - --- ------- ------- -------
ROW_MOVE GLO USE DURATION        SKIP_COR MON
-------- --- --- --------------- -------- ---
CLUSTER_OWNER
--------------------------------------------------------------------------------
DEPENDEN COMPRESS COMPRESS_FOR                   DRO REA SEG RESULT_ CLU
-------- -------- ------------------------------ --- --- --- ------- ---
ACTIVITY_TRACKING       DML_TIMESTAMP             HAS CON INMEMORY INMEMORY
----------------------- ------------------------- --- --- -------- --------
INMEMORY_DISTRI INMEMORY_COMPRESS INMEMORY_DUPL
--------------- ----------------- -------------
DEFAULT_COLLATION
--------------------------------------------------------------------------------
D S S E E EXT HYB CELLMEMORY               CON CON EXT EXT INMEMORY_SER
- - - - - --- --- ------------------------ --- --- --- --- ------------
INMEMORY_SERVICE_NAME
--------------------------------------------------------------------------------
CON MEMOPTIM MEMOPTIM HAS ADM DAT LOGICAL_ STA ROW HAS
--- -------- -------- --- --- --- -------- --- --- ---
VECTOR_INDEX_TYPE
-----------------------------
```

mengambil dan menampilkan informasi tabel EMP milik user ALEX yang diambil dari data dictionary view DBA_TABLES

SQL> SELECT * FROM DBA_SEGMENTS

WHERE OWNER='ALEX'

AND SEGMENT_NAME='EMP';

```
SQL> SELECT * FROM DBA_SEGMENTS
WHERE OWNER='ALEX'
AND SEGMENT_NAME='EMP';
  2    3
no rows selected
```

mengambil dan menampilkan informasi dari tabel EMP milik user ALEX. Hasil 'no rows selected' menunjukkan tabel tersebut belum ada data yang dimasukkan

--SO ORACLE DID NOT CREATE SEGMENT, NO ROWS IN THE TABLE

SQL> INSERT INTO ALEX.EMP VALUES (1,'KHALED','the best teacher ever :)');

SQL> commit;

```
SQL> INSERT INTO ALEX.EMP VALUES (1,'KHALED','the best teacher ever : )');

1 row created.

SQL> commit;

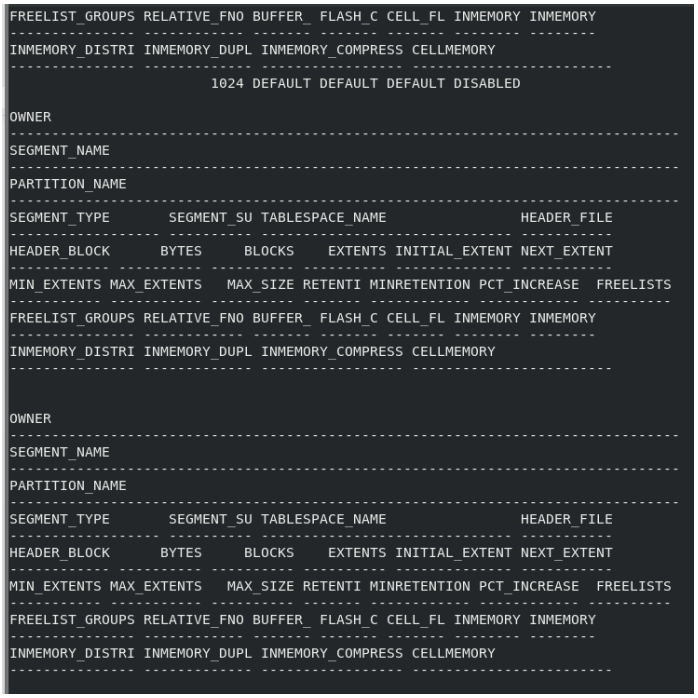
Commit complete.
```

Memasukkan satu baris data baru ke dalam tabel ALEX.EMP, dengan nilai EMP_ID adalah 1, NAME adalah 'KHALED', dan NOTE adalah 'the best teacher ever :)'. Lalu, commit untuk menyimpan perubahan data secara permanen

SQL> SELECT * FROM DBA_SEGMENTS

WHERE OWNER='ALEX'

AND SEGMENT_NAME='EMP';



mengambil dan menampilkan informasi dari tabel EMP milik user ALEX yang diambil dari data dictionary view DBA_SEGMENTS

SQL> select * from dba_extents

where owner='ALEX'

and SEGMENT_NAME='EMP';

```
HEADER_BLOCK      BYTES     BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT
------------ ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS   MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS
----------- ----------- ---------- ------- ------------ ------------ ----------
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ------------------ ------------------------


SQL> select * from dba_extents
where owner='ALEX'
and  SEGMENT_NAME='EMP';
  2    3
OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                 EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
ALEX
EMP


OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                 EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
TABLE              TS1                                     0         19
      8696      65536          8         1024
```

mengambil dan menampilkan informasi lengkap tabel EMP milik user ALEX yang diambil dari data dictionary view DBA_EXTENTS

--now let us insert 1000 rows

SQL> begin

for i in 1..1000

loop

insert into ALEX.EMP values (i,'just test name', 'just test comments');

end loop;

commit;

end;

```
SQL> SQL> begin
 for i in 1..1000
 loop
 insert into ALEX.EMP values (i,'just test name', 'just test comments');
 end loop;
 commit;
end;
  2    3    4    5    6    7    8  /

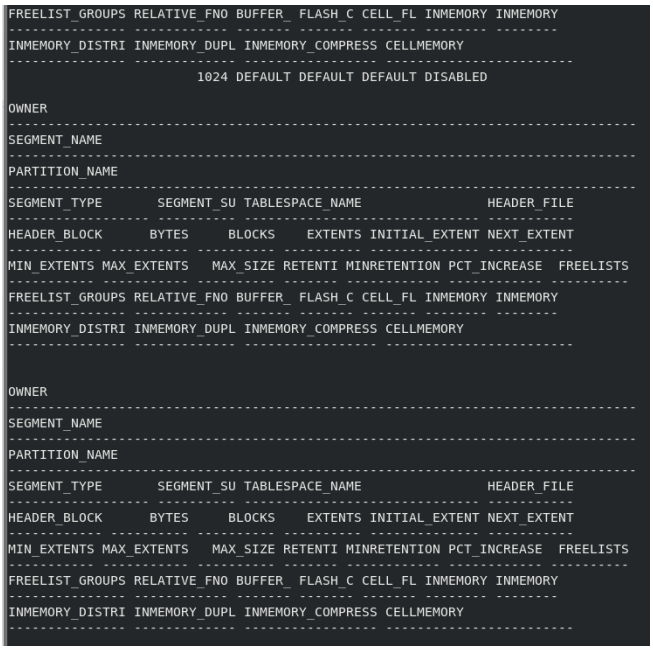
PL/SQL procedure successfully completed.
```

menjalankan PL/SQL untuk memasukkan (INSERT) data sebanyak 1000 baris ke dalam tabel ALEX.EMP menggunakan perulangan (loop). Setelah itu, melakukan commit untuk menyimpan perubahan data secara permanen

SQL> SELECT * FROM DBA_SEGMENTS

WHERE OWNER='ALEX'

AND SEGMENT_NAME='EMP';

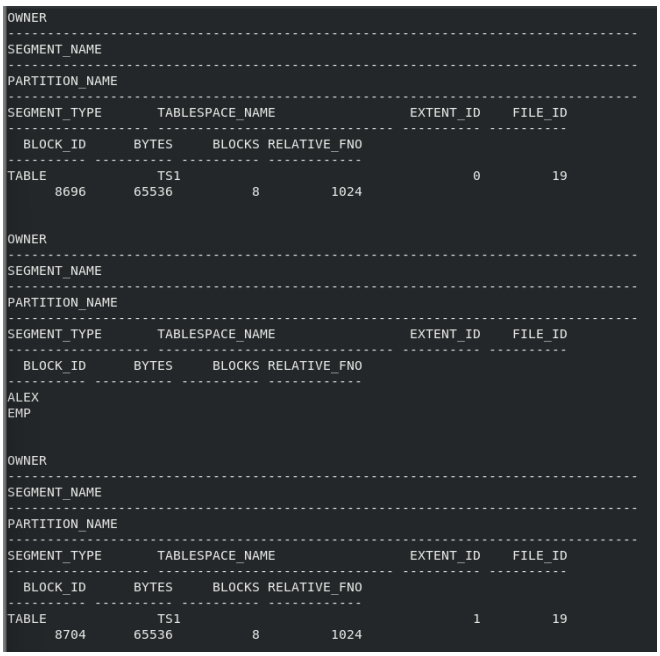


mengambil dan menampilkan kembali informasi dari tabel EMP milik user ALEX yang diambil dari data dictionary view DBA_SEGMENTS untuk melihat perubahan penyimpanan setelah penambahan data yang banyak

SQL> select * from dba_extents

where owner='ALEX'

and SEGMENT_NAME='EMP';



mengambil dan menampilkan kembali informasi tabel EMP milik user ALEX yang diambil dari data dictionary view DBA_EXTENTS, untuk melihat penambahan alokasi penyimpanan setelah input data massal

--now let us insert 1000000 rows

SQL> begin

for i in 1..1000000

loop

insert into ALEX.EMP values (i,'just test name', 'just test comments');

end loop;

commit;

end;

```
SQL> begin
 for i in 1..1000000
 loop
 insert into ALEX.EMP values (i,'just test name', 'just test comments');
 end loop;
 commit;
end;
  2    3    4    5    6    7    8  /

PL/SQL procedure successfully completed.
```

memasukkan (INSERT) data (1 juta baris) ke tabel ALEX.EMP menggunakan perulangan (loop). Lalu, commit untuk menyimpan perubahan secara permanen

SQL> SELECT * FROM DBA_SEGMENTS

WHERE OWNER='ALEX'

AND SEGMENT_NAME='EMP';

```
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ----------------- ------------------------
                        1024 DEFAULT DEFAULT DEFAULT DISABLED

OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                HEADER_FILE
------------------ ---------- ------------------------------ -----------
HEADER_BLOCK      BYTES     BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT
------------ ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS   MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS
----------- ----------- ---------- ------- ------------ ------------ ----------
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ----------------- ------------------------


OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                HEADER_FILE
------------------ ---------- ------------------------------ -----------
HEADER_BLOCK      BYTES     BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT
------------ ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS   MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS
----------- ----------- ---------- ------- ------------ ------------ ----------
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ----------------- ------------------------
```

mengambil dan menampilkan tabel EMP milik user ALEX yang diambil dari data dictionary view DBA_SEGMENTS, guna melihat pertumbuhan ukuran tabel (bytes/blocks) setelah penambahan 1000 baris data

SQL> select * from dba_extents

where owner='ALEX'

and SEGMENT_NAME='EMP';

```
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
TABLE              TS1                                 62         19
     14720    1048576        128         1024


OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
ALEX
EMP


OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
TABLE              TS1                                 63         19
     14848    1048576        128         1024


64 rows selected.
```

mengambil dan menampilkan informasi tentang ekstensi (extentsk tabel EMP milik user ALEX yang diambil dari data dictionary view DBA_EXTENTS, untuk memantau bagaimana data tersebar di berbagai penyimpanan

SQL> select rowid,emp_id,name,note

from ALEX.emp

where emp_id=1000000;

```
SQL> select rowid,emp_id,name,note
from ALEX.emp
where emp_id=1000000;
  2    3
ROWID                  EMP_ID
------------------ ----------
NAME
--------------------------------------------------------------------------------
NOTE
--------------------------------------------------------------------------------
AAAR8IAAAAAADpGABJ    1000000
just test name
just test comments
```

mengambil dan menampilkan data rowid (alamat fisik penyimpanan baris), emp_id, name, dan note dari tabel ALEX.EMP khusus untuk data dengan emp_id 1.000.000, memastikan insert data berhasil dilakukan sampai baris terakhir

--example

--rowid OOOOOOFFFBBBBBBRRR

--OOOOOO object number

--FFF Relative File Number

--BBBBBB The data block number

--RRR The row number

SQL> delete from ALEX.EMP;

SQL> commit;

```
SQL> delete from ALEX.EMP;

1001001 rows deleted.

SQL> commit;

Commit complete.
```

menghapus semua baris data dari tabel ALEX.EMP, kemudian lakukan commit untuk menyimpan dan mengakhiri transaksi, memastikan perubahan menjadi permanen

SQL> SELECT * FROM DBA_SEGMENTS

WHERE OWNER='ALEX'

AND SEGMENT_NAME='EMP';

```
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ----------------- ------------------------
                        1024 DEFAULT DEFAULT DEFAULT DISABLED

OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                HEADER_FILE
------------------ ---------- ------------------------------ -----------
HEADER_BLOCK      BYTES     BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT
------------ ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS   MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS
----------- ----------- ---------- ------- ------------ ------------ ----------
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ----------------- ------------------------


OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                HEADER_FILE
------------------ ---------- ------------------------------ -----------
HEADER_BLOCK      BYTES     BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT
------------ ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS   MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS
----------- ----------- ---------- ------- ------------ ------------ ----------
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ----------------- ------------------------
```

SQL> select * from dba_extents

where owner='ALEX'

and SEGMENT_NAME='EMP';

```
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
TABLE              TS1                                    62         19
     14720    1048576        128         1024


OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
ALEX
EMP


OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
TABLE              TS1                                    63         19
     14848    1048576        128         1024


64 rows selected.
```

mengambil dan menampilkan kembali informasi lengkap tentang ekstensi (extents) atau alokasi penyimpanan fisik untuk tabel EMP user ALEX dari view DBA_EXTENTS. Tujuannya untuk memastikan bahwa meskipun data telah dihapus (DELETE), extents biasanya masih tetap teralokasi dan tidak langsung dilepaskan kembali ke database

SQL> truncate table ALEX.EMP;

```
SQL> truncate table ALEX.EMP;

Table truncated.
```

mengosongkan (TRUNCATE) tabel ALEX.EMP dengan menghapus semua baris data

SQL> SELECT * FROM DBA_SEGMENTS

WHERE OWNER='ALEX'

AND SEGMENT_NAME='EMP';

```
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ----------------- ------------------------
                         1024 DEFAULT DEFAULT DEFAULT DISABLED

OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                HEADER_FILE
------------------ ---------- ------------------------------ -----------
HEADER_BLOCK      BYTES     BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT
------------ ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS   MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS
----------- ----------- ---------- ------- ------------ ------------ ----------
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ----------------- ------------------------

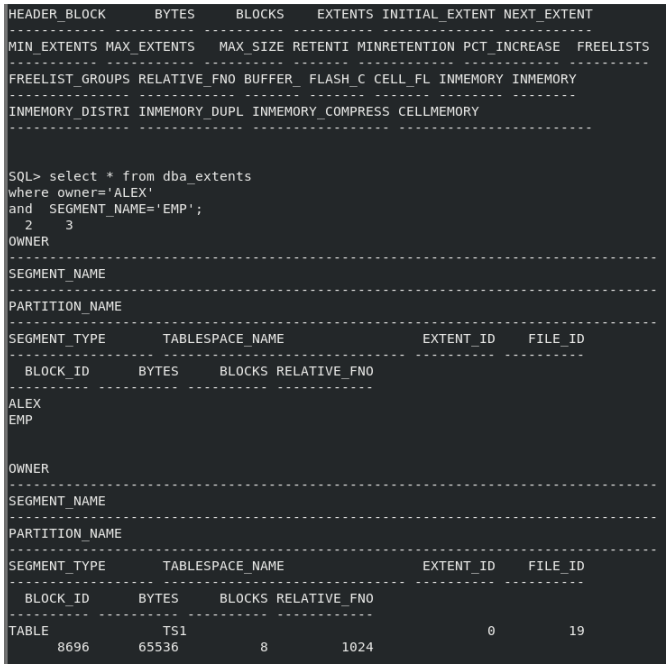

OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                HEADER_FILE
------------------ ---------- ------------------------------ -----------
HEADER_BLOCK      BYTES     BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT
------------ ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS   MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS
----------- ----------- ---------- ------- ------------ ------------ ----------
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ----------------- ------------------------
```

mengambil dan menampilkan kembali informasi tabel EMP user ALEX yang diambil dari data dictionary view DBA_SEGMENTS, memastikan bahwa setelah perintah TRUNCATE, bytes/blocks dilepaskan dan kembali ke ukuran awal (reset), berbeda dengan perintah DELETE yang tidak melepaskan penyimpanan

SQL> select * from dba_extents

where owner='ALEX'

and SEGMENT_NAME='EMP';



```
HEADER_BLOCK      BYTES     BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT
------------ ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS   MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS
----------- ----------- ---------- ------- ------------ ------------ ----------
FREELIST_GROUPS RELATIVE_FNO BUFFER_ FLASH_C CELL_FL INMEMORY INMEMORY
--------------- ------------ ------- ------- ------- -------- --------
INMEMORY_DISTRI INMEMORY_DUPL INMEMORY_COMPRESS CELLMEMORY
--------------- ------------- ------------------ ------------------------


SQL> select * from dba_extents
where owner='ALEX'
and  SEGMENT_NAME='EMP';
  2    3
OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
ALEX
EMP

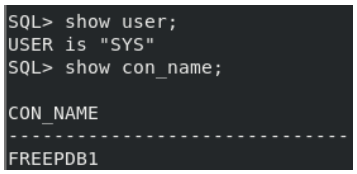

OWNER
--------------------------------------------------------------------------------
SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       TABLESPACE_NAME                EXTENT_ID    FILE_ID
------------------ ------------------------------ ---------- ----------
  BLOCK_ID      BYTES     BLOCKS RELATIVE_FNO
---------- ---------- ---------- ------------
TABLE              TS1                                     0         19
      8696      65536          8         1024
```

mengambil dan menampilkan kembali informasi extents dari tabel EMP user ALEX yang diambil dari view DBA_EXTENTS, memastikan setelah perintah TRUNCATE, extents benar-benar telah dilepaskan dan dikembalikan ke database (biasanya hanya menyisakan extent awal saja)

==========

SQL> show user;

SQL> show con_name;



```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

show user;: Menampilkan user yang sedang terhubung

show con_name;: Menampilkan user saat ini terhubung ke PDB/ CDB

SQL> create table dept

(deptno NUMBER(2),

dept_name varchar2(100)

);

```
SQL> create table dept
(deptno NUMBER(2),
 dept_name varchar2(100 )
 );
  2    3    4
Table created.
```

membuat tabel baru bernama dept. Tabel ini memiliki dua kolom: deptno (NUMBER(2)) dan dept_name (VARCHAR2(100))

SQL> select * from dba_tables

where table_name='DEPT';

```
OWNER
------------------------------------------------------------------------------
TABLE_NAME
------------------------------------------------------------------------------
TABLESPACE_NAME
------------------------------
CLUSTER_NAME
------------------------------------------------------------------------------
IOT_NAME
------------------------------------------------------------------------------
STATUS     PCT_FREE   PCT_USED  INI_TRANS  MAX_TRANS INITIAL_EXTENT NEXT_EXTENT
-------- ---------- ---------- ---------- ---------- -------------- -----------
MIN_EXTENTS MAX_EXTENTS PCT_INCREASE  FREELISTS FREELIST_GROUPS LOG B   NUM_ROWS
----------- ----------- ------------ ---------- --------------- --- - ----------
    BLOCKS EMPTY_BLOCKS  AVG_SPACE  CHAIN_CNT AVG_ROW_LEN
---------- ------------ ---------- ---------- -----------
AVG_SPACE_FREELIST_BLOCKS NUM_FREELIST_BLOCKS DEGREE     INSTANCES  CACHE
------------------------- -------------------- ---------- ---------- -----
TABLE_LO SAMPLE_SIZE LAST_ANAL PAR IOT_TYPE     T S NES BUFFER_ FLASH_C CELL_FL
-------- ----------- ---------- --- ------------ - - --- ------- ------- -------
ROW_MOVE GLO USE DURATION        SKIP_COR MON
-------- --- --- --------------- -------- ---
CLUSTER_OWNER
------------------------------------------------------------------------------
DEPENDEN COMPRESS COMPRESS_FOR                   DRO REA SEG RESULT_ CLU
-------- -------- ------------------------------ --- --- --- ------- ---
ACTIVITY_TRACKING       DML_TIMESTAMP             HAS CON INMEMORY INMEMORY
----------------------- ------------------------- --- --- -------- --------
INMEMORY_DISTRI INMEMORY_COMPRESS INMEMORY_DUPL
--------------- ----------------- -------------
DEFAULT_COLLATION
------------------------------------------------------------------------------
D S S E E EXT HYB CELLMEMORY               CON CON EXT EXT INMEMORY_SER
- - - - - --- --- ------------------------ --- --- --- --- ------------
INMEMORY_SERVICE_NAME
------------------------------------------------------------------------------
CON MEMOPTIM MEMOPTIM HAS ADM DAT LOGICAL_ STA ROW HAS
--- -------- -------- --- --- --- -------- --- --- ---
VECTOR_INDEX_TYPE
---------------------------
```

mengambil dan menampilkan informasi tentang tabel bernama DEPT yang diambil dari data dictionary view DBA_TABLES

SQL> SELECT * FROM USER_SEGMENTS

WHERE SEGMENT_NAME='DEPT';

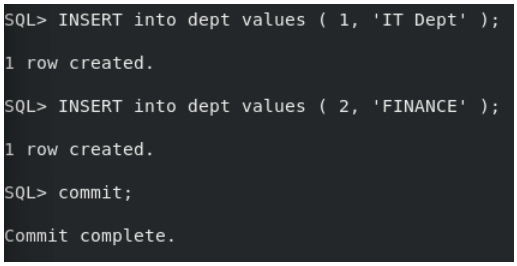


mengambil dan menampilkan informasi dari tabel DEPT yang diambil dari data dictionary view USER_SEGMENTS

SQL> INSERT into dept values (1, 'IT Dept');

SQL> INSERT into dept values (2, 'FINANCE');

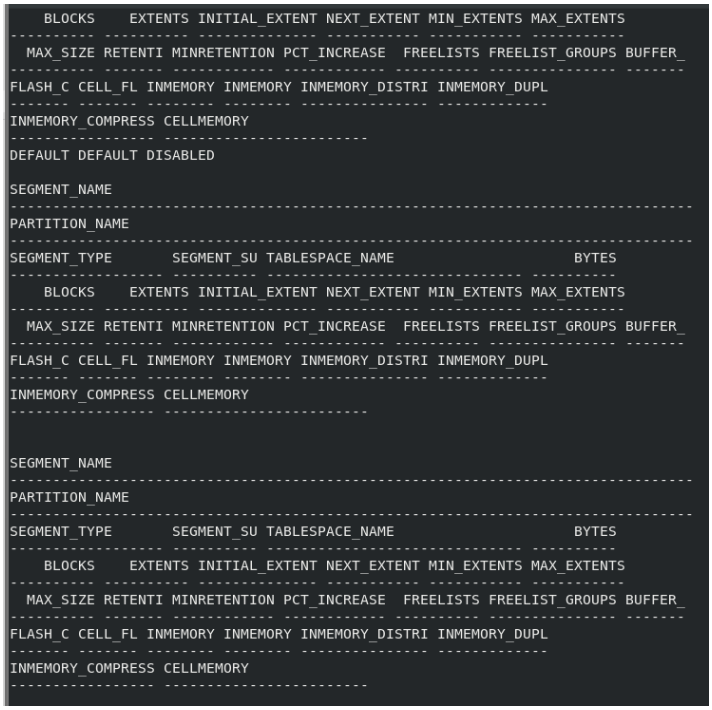
SQL> commit;



memasukkan dua baris data baru ke dalam tabel dept. Baris pertama dengan nilai 1 dan 'IT Dept', sedangkan baris kedua dengan nilai 2 dan 'FINANCE'. Kemudian, melakukan commit untuk menyimpan perubahan data secara permanen

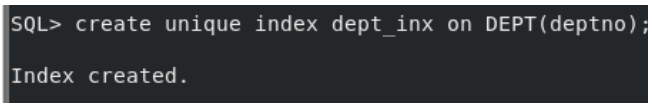
SQL> SELECT * FROM USER_SEGMENTS

WHERE SEGMENT_NAME='DEPT';



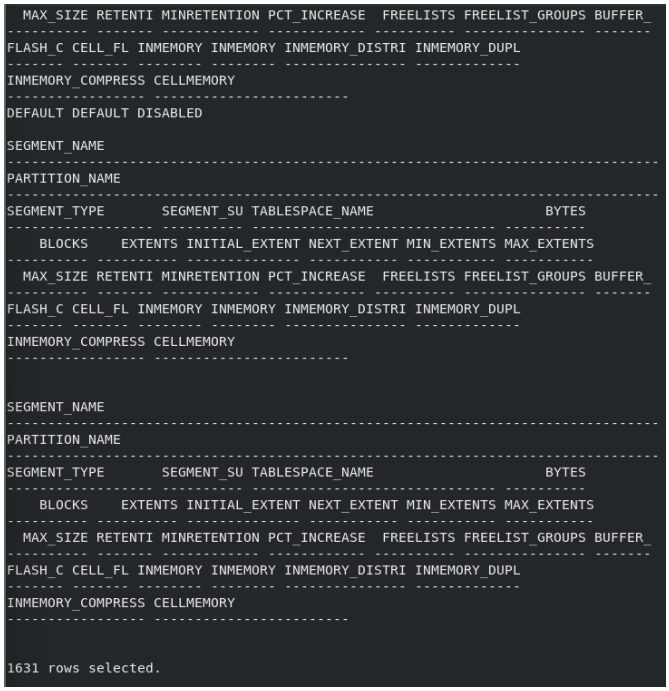
mengambil dan menampilkan kembali informasi dari tabel DEPT yang diambil dari data dictionary view USER_SEGMENTS

SQL> create unique index dept_inx on DEPT(deptno);



membuat unique index (indeks unik) baru bernama dept_inx pada tabel DEPT untuk kolom dept

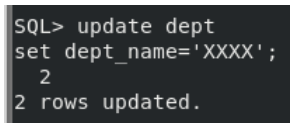
SQL> SELECT * FROM USER_SEGMENTS;



mengambil dan menampilkan semua informasi yang dimiliki oleh user saat ini, yang diambil dari data dictionary view USER_SEGMENTS

SQL> update dept

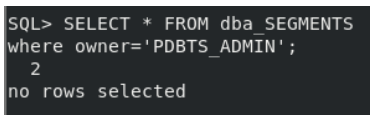
set dept_name='XXXX';



mengubah (UPDATE) data pada kolom dept_name menjadi 'XXXX' di dalam tabel DEPT

SQL> SELECT * FROM dba_SEGMENTS

where owner='PDBTS_ADMIN';



mengambil dan menampilkan semua informasi yang dimiliki user PDBTS_ADMIN yang diambil dari data dictionary view DBA_SEGMENTS

SQL> select a.sid, a.serial#, a.username, b.used_urec, b.used_ublk

from v$session a,

v$transaction b

where a.saddr = b.ses_addr

order by b.used_ublk desc;

```
SQL> SELECT * FROM dba_SEGMENTS
where owner='PDBTS_ADMIN';
  2
no rows selected

SQL> select a.sid, a.serial#, a.username, b.used_urec, b.used_ublk
from   v$session a,
       v$transaction b
where  a.saddr = b.ses_addr
order by b.used_ublk desc;
  2    3    4    5
       SID    SERIAL#
---------- ----------
USERNAME
--------------------------------------------------------------------------------
 USED_UREC  USED_UBLK
---------- ----------
        16      20923
SYS
         2          1
```

mengambil dan menampilkan informasi transaksi dengan menggabungkan view dynamic performance v$session dan v$transaction

--USED_UBLK = Number of undo blocks used

--USED_UREC = Number of undo records used

SQL> ROLLBACK;

```
SQL> ROLLBACK;

Rollback complete.
```

membatalkan (ROLLBACK) semua perubahan data yang dilakukan dalam transaksi saat ini yang belum disimpan secara permanen (belum di commit), mengembalikan ke kondisi sebelum transaksi dimulai

SQL> select a.sid, a.serial#, a.username, b.used_urec, b.used_ublk

from   v$session a,

      v$transaction b

where a.saddr = b.ses_addr

order by b.used_ublk desc;

```
SQL> select a.sid, a.serial#, a.username, b.used_urec, b.used_ublk
from   v$session a,
       v$transaction b
where  a.saddr = b.ses_addr
order by b.used_ublk desc;
  2    3    4    5
no rows selected
```

mengambil dan menampilkan informasi tentang transaksi yang berjalan dengan menggabungkan view v$session dan v$transaction

==========

--DEFERRED_SEGMENT_CREATION

SQL> show user;

SQL> show con_name;

```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

show user;: Menampilkan user yang sedang terhubung

show con_name;: Menampilkan user terhubung ke PDB/ CDB

SQL> show parameter DEFERRED_SEGMENT_CREATION;

```
SQL> show parameter DEFERRED_SEGMENT_CREATION;

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
deferred_segment_creation            boolean     TRUE
```

menampilkan pengaturan parameter DEFERRED_SEGMENT_CREATION. Parameter ini menentukan apakah pembuatan segmen fisik (ruang penyimpanan) akan ditunda sampai data pertama dimasukkan (TRUE) atau langsung dibuat saat tabel dibuat (FALSE)

--this mean the segment will created for the table

--1 after you crate the table using DDL statement (create table)

--2 after you insert one record (DML)

SQL> create table country (id number, name varchar2(100));

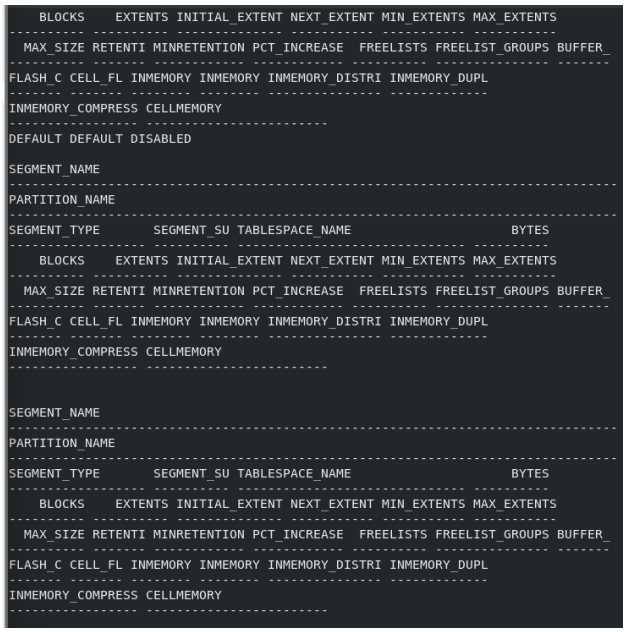
```
SQL> create table country (id number, name varchar2(100) );

Table created.
```

membuat tabel baru bernama COUNTRY dengan dua kolom: ID (tipe data NUMBER) dan NAME (tipe data VARCHAR2 dengan panjang maksimal 100 karakter)

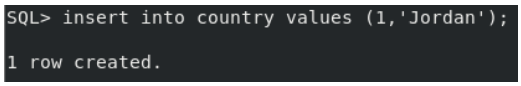
SQL> select * from user_segments

where segment_name= upper('country');



mengambil dan menampilkan informasi tabel COUNTRY yang diambil dari data dictionary view USER_SEGMENTS

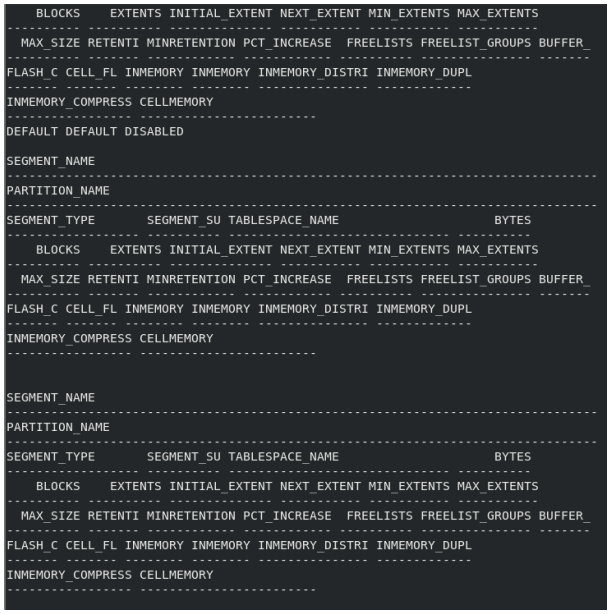
SQL> insert into country values (1,'Jordan');



memasukkan satu baris data baru ke dalam tabel COUNTRY, dengan nilai id adalah 1 dan name adalah 'Jordan'

SQL> select * from user_segments

where segment_name= upper('country');



mengambil dan menampilkan kembali informasi dari tabel COUNTRY yang diambil dari data dictionary view USER_SEGMENTS, untuk memantau status alokasi penyimpanan setelah penambahan data

--now let change DEFERRED_SEGMENT_CREATION to be false

--you can do this by alter session / alter system

--this will create the segment immediate

SQL> alter session set DEFERRED_SEGMENT_CREATION=false;

```
SQL> alter session set DEFERRED_SEGMENT_CREATION=false;

Session altered.
```

mengubah pengaturan sesi saat ini (ALTER SESSION) untuk menonaktifkan fitur DEFERRED_SEGMENT_CREATION

SQL> show parameter DEFERRED_SEGMENT_CREATION;

```
SQL> show parameter DEFERRED_SEGMENT_CREATION;

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
deferred_segment_creation            boolean     FALSE
```

menampilkan nilai parameter DEFERRED_SEGMENT_CREATION untuk memastikan pengaturannya saat ini, apakah bernilai TRUE (pembuatan segmen ditunda sampai ada data) atau FALSE (segmen langsung dibuat saat tabel didefinisikan)

SQL> create table jobs (id number, title varchar2(100));

```
SQL> create table jobs (id number, title varchar2(100) );

Table created.
```

membuat tabel baru bernama JOBS dengan dua kolom: id (tipe data NUMBER) dan title (tipe data VARCHAR2 dengan panjang maksimal 100 karakter)

SQL> select * from user_segments

where segment_name= upper('jobs');

```
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------
DEFAULT DEFAULT DISABLED

SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                     BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------


SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                     BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------
```

mengambil dan menampilkan informasi dari tabel JOBS yang diambil dari data dictionary view USER_SEGMENTS

--note

--Deferred segment is the default for tables, indexes, and partitions

SQL> alter session set DEFERRED_SEGMENT_CREATION=true;

```
SQL> alter session set DEFERRED_SEGMENT_CREATION=true;

Session altered.
```

mengubah pengaturan sesi (ALTER SESSION) untuk mengaktifkan kembali fitur DEFERRED_SEGMENT_CREATION (set ke TRUE)

--but we can do this also

SQL> create table test (n number)

SEGMENT CREATION IMMEDIATE;

```
SQL> create table test ( n number )
SEGMENT CREATION IMMEDIATE;
  2
Table created.
```

membuat tabel baru bernama TEST dengan satu kolom n (NUMBER). Opsi SEGMENT CREATION IMMEDIATE memaksa database untuk langsung mengalokasikan ruang penyimpanan fisik (segmen) saat tabel dibuat, tanpa menunda hingga ada data yang masuk

SQL> select * from user_segments

where segment_name= upper('test');

```
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
---------------- ------------------------
DEFAULT DEFAULT DISABLED

SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                     BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
---------------- ------------------------


SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                     BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
---------------- ------------------------
```

mengambil dan menampilkan informasi dari tabel TEST yang diambil dari data dictionary view USER_SEGMENTS

SQL> create table test10 (n number)

SEGMENT CREATION DEFERRED;

```
SQL> create table test10 ( n number )
SEGMENT CREATION DEFERRED;
  2  create table test10 ( n number )
*
ERROR at line 1:
ORA-14224: Deferred segment creation is not supported for this schema
Help: https://docs.oracle.com/error-help/db/ora-14224/
```

mencoba membuat tabel bernama TEST10 dengan opsi SEGMENT CREATION DEFERRED

SQL> select * from user_segments

where segment_name= upper('test10');

```
SQL> select * from user_segments
where segment_name= upper('test10');
  2
no rows selected
```

mengambil dan menampilkan informasi dari tabel TEST10 yang diambil dari data dictionary view USER_SEGMENTS. Hasilnya no rows selected karena sebelumnya error, sehingga tabel TEST10 sebenarnya tidak pernah terbentuk

==========

--Space-Saving Features (UNUSABLE index)

SQL> show user;

SQL> show con_name;

```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

show user;: Menampilkan user yang sedang terhubung

show con_name;: Menampilkan user sekarang terhubung ke PDB/ CDB

SQL> show parameter DEFERRED_SEGMENT_CREATION;

```
SQL> show parameter DEFERRED_SEGMENT_CREATION;

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
deferred_segment_creation            boolean     TRUE
```

menampilkan nilai parameter DEFERRED_SEGMENT_CREATION. Parameter ini menentukan apakah akan ditunda sampai data pertama dimasukkan (TRUE) atau langsung dibuat saat tabel didefinisikan (FALSE)

SQL> create table employees

(emp_id number,

fname varchar2(100),

lname varchar2(100),

constraint employees_pk primary key (emp_id)

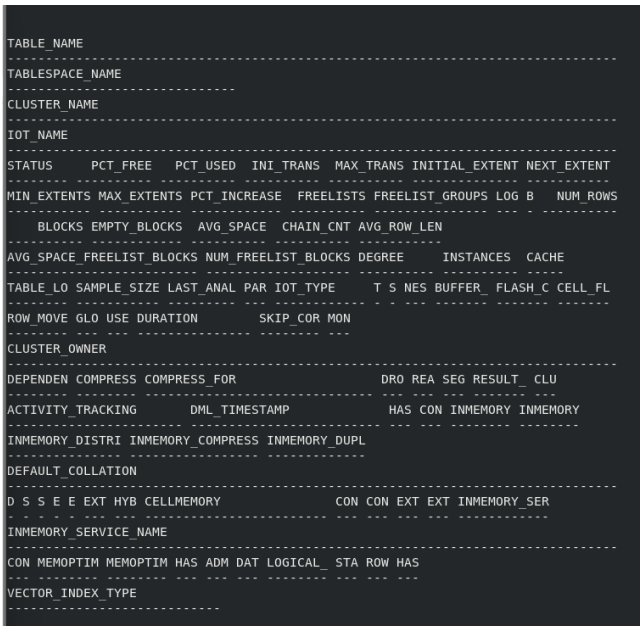
);

```
SQL> create table employees
( emp_id number,
  fname varchar2(100),
  lname varchar2(100),
 constraint employees_pk primary key (emp_id)
 );
  2    3    4    5    6
Table created.
```

membuat tabel baru bernama EMPLOYEES. Tabel ini memiliki tiga kolom: EMP_ID (NUMBER), FNAME (VARCHAR2(100)), dan LNAME (VARCHAR2(100)). Kolom EMP_ID ditetapkan sebagai Primary Key melalui constraint EMPLOYEES_PK

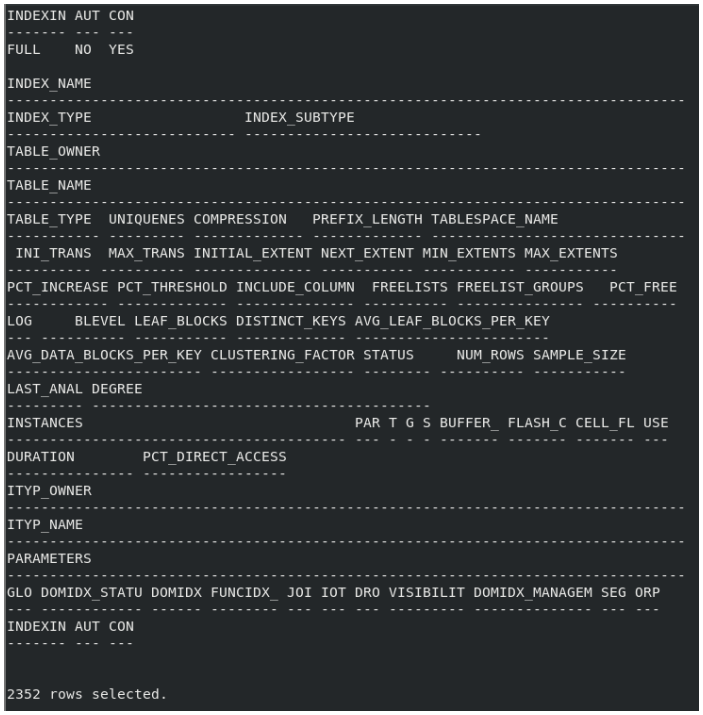
SQL> select * from user_tables

where table_name='EMPLOYEES';



mengambil dan menampilkan informasi lengkap tentang tabel EMPLOYEES yang diambil dari data dictionary view USER_TABLES

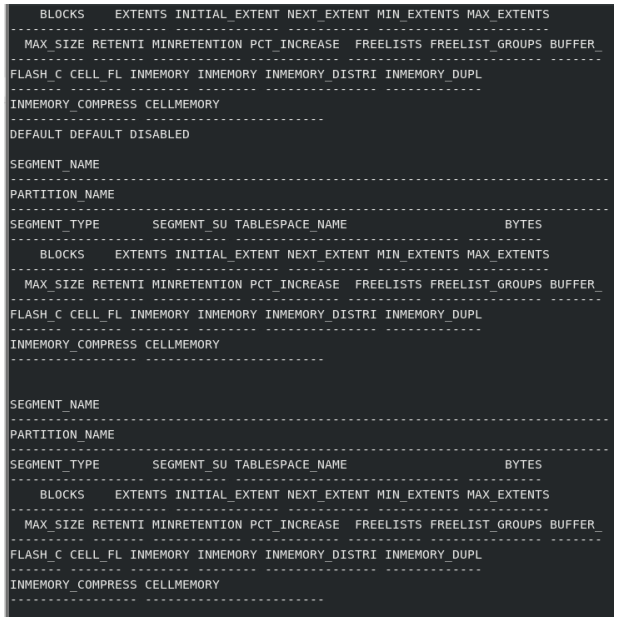
SQL> SELECT * FROM USER_INDEXES;



mengambil dan menampilkan semua informasi lengkap tentang indeks (indexes) yang dimiliki oleh user sekarang, yang diambil dari data dictionary view USER_INDEXES

SQL> select * from user_segments

where segment_name IN ('EMPLOYEES','EMPLOYEES_PK');



mengambil dan menampilkan informasi tabel EMPLOYEES dan indeks EMPLOYEES_PK yang diambil dari data dictionary view USER_SEGMENTS, untuk memeriksa apakah ruang penyimpanan sudah dialokasikan untuk tabel dan indeks primary keynya

SQL> insert into employees values (1,'khaled','alkhudari');

SQL> commit;

```
SQL> insert into employees values (1,'khaled','alkhudari');

1 row created.

SQL> commit;

Commit complete.
```

memasukkan (INSERT) satu baris data baru ke dalam tabel EMPLOYEES, dengan nilai emp_id adalah 1, fname adalah 'khaled', dan lname adalah 'alkhudari'. Kemudian, commit untuk menyimpan perubahan data secara permanen

SQL> select * from employees;

```
SQL> select * from employees;

    EMP_ID
----------
FNAME
--------------------------------------------------------------------------------
LNAME
--------------------------------------------------------------------------------
         1
khaled
alkhudari
```

mengambil dan menampilkan semua data dari tabel EMPLOYEES

SQL> select * from user_segments

where segment_name IN ('EMPLOYEES','EMPLOYEES_PK');

```
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------
DEFAULT DEFAULT DISABLED

SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                     BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------


SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                     BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------
```

mengambil dan menampilkan informasi tabel EMPLOYEES dan indeks EMPLOYEES_PK yang diambil dari data dictionary view USER_SEGMENTS

SQL> create index fname_index on EMPLOYEES(fname) ;

```
SQL> create index fname_index on EMPLOYEES(fname) ;

Index created.
```

membuat index baru yaitu fname_index pada tabel EMPLOYEES untuk kolom fname

SQL> select * from user_segments

where segment_name IN ('EMPLOYEES','EMPLOYEES_PK' ,'FNAME_INDEX')

```
---------- ---------- ------------- ----------- ---------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------
DEFAULT DEFAULT DISABLED

SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                    BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------


SEGMENT_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
SEGMENT_TYPE       SEGMENT_SU TABLESPACE_NAME                    BYTES
------------------ ---------- ------------------------------ ----------
    BLOCKS    EXTENTS INITIAL_EXTENT NEXT_EXTENT MIN_EXTENTS MAX_EXTENTS
---------- ---------- -------------- ----------- ----------- -----------
  MAX_SIZE RETENTI MINRETENTION PCT_INCREASE  FREELISTS FREELIST_GROUPS BUFFER_
---------- ------- ------------ ------------ ---------- --------------- -------
FLASH_C CELL_FL INMEMORY INMEMORY INMEMORY_DISTRI INMEMORY_DUPL
------- ------- -------- -------- --------------- -------------
INMEMORY_COMPRESS CELLMEMORY
----------------- ------------------------
```

mengambil dan menampilkan informasi tabel EMPLOYEES, indeks EMPLOYEES_PK, dan indeks FNAME_INDEX yang diambil dari data dictionary view USER_SEGMENTS

SQL> create index lname_index on EMPLOYEES(lname) UNUSABLE ;

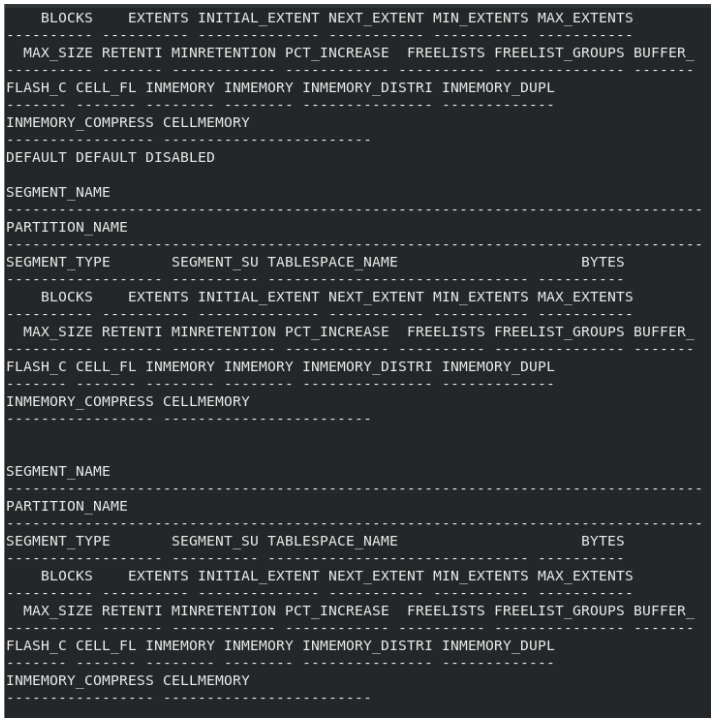
```
SQL> create index lname_index on EMPLOYEES(lname) UNUSABLE ;

Index created.
```

membuat index baru bernama lname_index pada tabel EMPLOYEES untuk kolom lname dengan status UNUSABLE

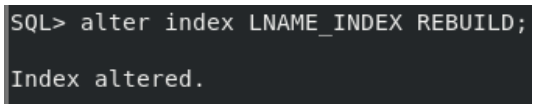
SQL> select * from user_segments

where segment_name IN ('EMPLOYEES','EMPLOYEES_PK' ,'FNAME_INDEX','LNAME_INDEX');



mengambil dan menampilkan informasi untuk tabel EMPLOYEES serta indeks EMPLOYEES_PK, FNAME_INDEX, dan LNAME_INDEX yang diambil dari data dictionary view USER_SEGMENTS

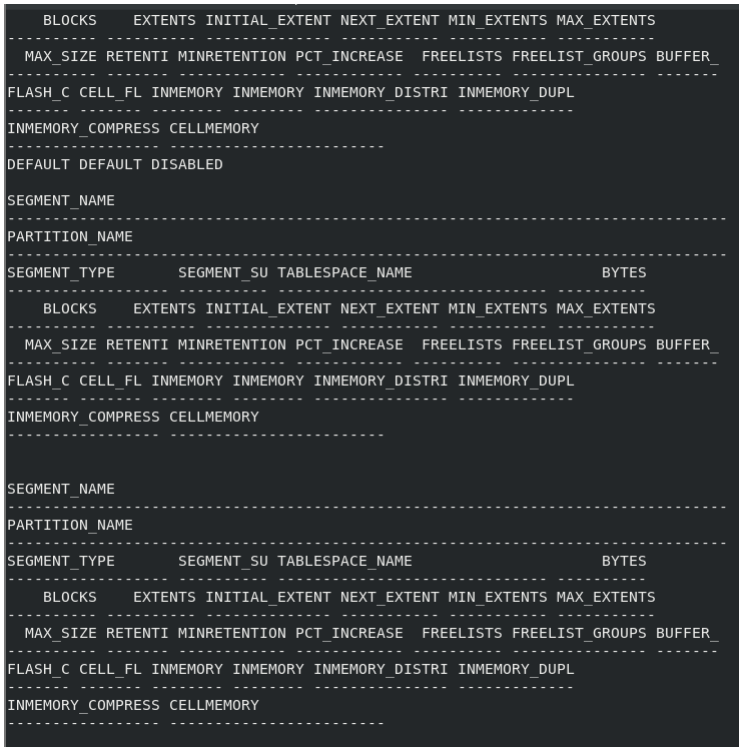
SQL> alter index LNAME_INDEX REBUILD;



membangun ulang (REBUILD) index bernama LNAME_INDEX, perintah ini akan mengalokasikan ruang penyimpanan fisik (segmen) yang sebelumnya tidak ada (karena statusnya UNUSABLE) dan mengaktifkan index agar dapat digunakan

SQL> select * from user_segments

where segment_name IN ('EMPLOYEES','EMPLOYEES_PK' ,'FNAME_INDEX','LNAME_INDEX');



mengambil dan menampilkan informasi tabel EMPLOYEES serta indeks EMPLOYEES_PK, FNAME_INDEX, dan LNAME_INDEX yang diambil dari data dictionary view USER_SEGMENTS

==========

--basic compression

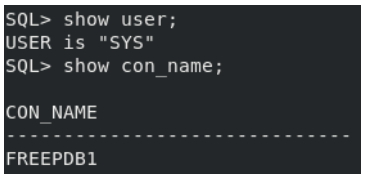
--With basic compression, the Oracle Database server compresses data at the time of performing bulk load using

--operations such as direct loads or CREATE TABLE AS SELECT

--Is recommended for bulk loading data warehouses

SQL> show user;

SQL> show con_name;

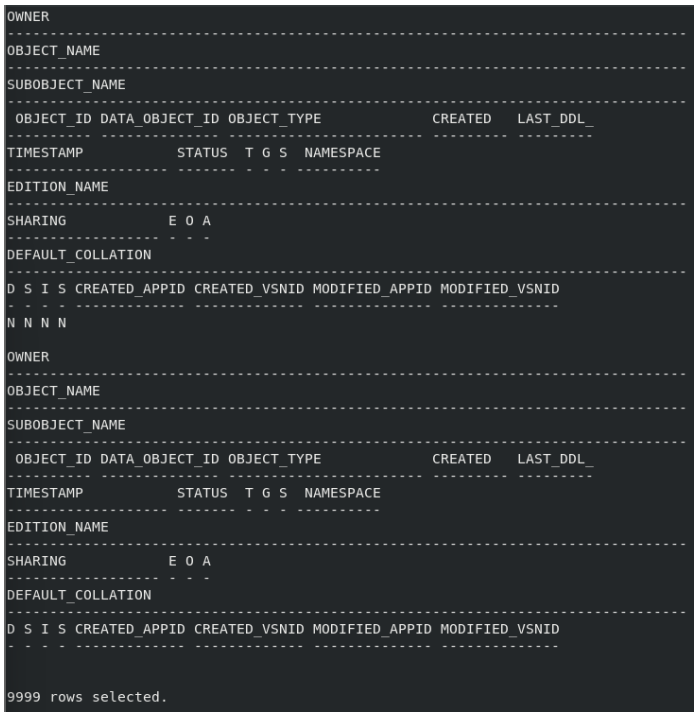


show user;: Menampilkan user yang sedang terhubung

show con_name;: Menampilkan user saat ini terhubung ke PDB / CDB

SQL> select * from dba_objects

where rownum<10000;



mengambil dan menampilkan semua informasi yang diambil dari data dictionary view DBA_OBJECTS, dengan batasan hanya menampilkan 9.999 baris pertama saja (kurang dari 10.000)

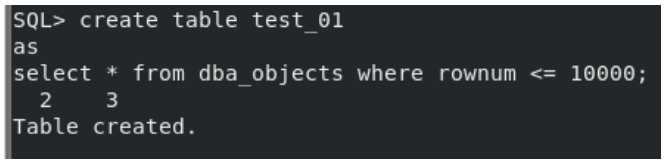
--case 1

--no compression

SQL> create table test_01

as

select * from dba_objects where rownum <= 10000;



membuat tabel baru bernama TEST_01 dengan menyalin struktur dan data dari data dictionary view DBA_OBJECTS, namun dibatasi hanya mengambil 10.000 baris data pertama

SQL> SELECT DBMS_METADATA.GET_DDL('TABLE','TEST_01') FROM dual;

```
SQL> SELECT DBMS_METADATA.GET_DDL('TABLE','TEST_01') FROM dual;

DBMS_METADATA.GET_DDL('TABLE','TEST_01')
--------------------------------------------------------------------------------

  CREATE TABLE "SYS"."TEST_01"
   (    "OWNER" VARCHAR2(128),
        "OBJECT_NAME" VAR
```

mengambil dan menampilkan kode DDL pembuatan tabel 'TEST_01' untuk melihat struktur dan konfigurasi lengkapnya secara detail

SQL> ANALYZE TABLE TEST_01 COMPUTE statistics;

```
SQL> ANALYZE TABLE TEST_01 COMPUTE statistics;

Table analyzed.
```

melakukan analisis dan perhitungan statistik lengkap (COMPUTE) pada tabel TEST_01

SQL> select blocks, pct_free , compression, compress_for

from   user_tables

where   table_name = 'TEST_01';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_01';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
       180         10 DISABLED
```

mengambil dan menampilkan informasi tentang penggunaan blok penyimpanan (blocks), persentase ruang kosong untuk update (pct_free), serta status dan jenis kompresi data (compression dan compress_for) pada tabel TEST_01 yang diambil dari data dictionary view USER_TABLES

--CASE 2

--with basic compression enabled

SQL> create table test_02 compress basic

as

select * from dba_objects where rownum <= 10000;

```
SQL> create table test_02 compress basic
as
select * from dba_objects where rownum <= 10000;
  2    3
Table created.
```

membuat tabel baru bernama TEST_02 dengan menyalin 10.000 baris data pertama dari view DBA_OBJECTS. Tabel ini menggunakan opsi COMPRESS BASIC, yang mengaktifkan kompresi data standar untuk menghemat penggunaan ruang penyimpanan (disk space)

SQL> ANALYZE TABLE TEST_02 COMPUTE statistics;

```
SQL> ANALYZE TABLE TEST_02 COMPUTE statistics;

Table analyzed.
```

melakukan analisis dan perhitungan statistik lengkap (COMPUTE) pada tabel TEST_02

SQL> select blocks, pct_free , compression, compress_for

from user_tables

where table_name = 'TEST_02';

```
SQL> select  blocks, pct_free , compression, compress_for
from  user_tables
where  table_name = 'TEST_02';
  2  3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
        37          0 ENABLED  BASIC
```

mengambil dan menampilkan informasi mengenai penggunaan blok penyimpanan (blocks), persentase ruang kosong (pct_free), serta status kompresi (compression dan compress_for) pada tabel TEST_02 yang diambil dari data dictionary view USER_TABLES

--CASE 3

--Normal insert into empty table defined as compress basic (no effects)

SQL> create table test_03 compress basic

as

select * from dba_objects where 1=2;

```
SQL> create table test_03 compress basic
as
select * from dba_objects where 1=2;
  2  3
Table created.
```

membuat tabel baru bernama TEST_03 dengan mengaktifkan opsi COMPRESS BASIC. Tabel ini hanya menyalin struktur kolom dari view DBA_OBJECTS tanpa menyalin datanya (tabel kosong), karena kondisi WHERE 1=2 memastikan tidak ada baris data yang diambil

SQL> select * from test_03;

```
SQL> select * from test_03;

no rows selected
```

mengambil dan menampilkan semua data dari tabel TEST_03

SQL> insert into test_03

select * from dba_objects where rownum <= 10000;

commit;

```
SQL> insert into test_03
select * from dba_objects where rownum <= 10000;
commit;
  2
10000 rows created.

SQL>
Commit complete.
```

memasukkan (INSERT) 10.000 baris data pertama dari data dictionary view DBA_OBJECTS ke dalam tabel TEST_03, kemudian melakukan commit untuk menyimpan data tersebut secara permanen

SQL> select blocks, pct_free , compression, compress_for

from   user_tables

where   table_name = 'TEST_03';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_03';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
                    0 ENABLED  BASIC
```

mengambil dan menampilkan informasi penggunaan blok penyimpanan (blocks), persentase ruang kosong (pct_free), serta status dan jenis kompresi (compression, compress_for) pada tabel TEST_03 yang diambil dari data dictionary view USER_TABLES

SQL> ANALYZE TABLE TEST_03 COMPUTE statistics;

```
SQL> ANALYZE TABLE TEST_03 COMPUTE statistics;

Table analyzed.
```

melakukan analisis dan perhitungan statistik lengkap (COMPUTE) pada tabel TEST_03

SQL> select blocks, pct_free , compression, compress_for

from   user_tables

where   table_name = 'TEST_03';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_03';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
       162          0 ENABLED  BASIC
```

mengambil dan menampilkan informasi tentang penggunaan blok penyimpanan (blocks), persentase ruang kosong (pct_free), serta status dan jenis kompresi (compression, compress_for) pada tabel TEST_03 yang diambil dari data dictionary view USER_TABLES

--4. Direct path insert into empty table defined as compressed

SQL> create table TEST_04 compress basic

as

select * from dba_objects where rownum = 0;

```
SQL> create table TEST_04 compress basic
as
select * from dba_objects where rownum = 0;
  2    3
Table created.
```

membuat tabel baru bernama TEST_04 dengan mengaktifkan opsi COMPRESS BASIC. Tabel ini hanya menyalin struktur kolom dari view DBA_OBJECTS tanpa menyalin datanya (tabel kosong), karena kondisi WHERE rownum = 0 memastikan tidak ada baris data yang diambil

SQL> select * from TEST_04;

```
SQL> select * from TEST_04;

no rows selected
```

mengambil dan menampilkan semua data dari tabel TEST_04. Hasilnya **'no rows selected'**, karena pada langkah sebelumnya tabel ini dibuat dengan kondisi WHERE rownum = 0, yang artinya hanya struktur tabelnya saja yang disalin tanpa membawa baris datanya

SQL> insert /*+ append */ into TEST_04 select * from dba_objects where rownum <= 10000;

```
SQL> insert /*+ append */ into TEST_04 select * from dba_objects where rownum <= 10000;

10000 rows created.
```

memasukkan (INSERT) 10.000 baris data ke dalam tabel TEST_04 menggunakan hint /*+ append */

/*

The APPEND hint tells the optimizer to perform a direct-path insert, which improves the

performance of INSERT .. SELECT operations for a number of reasons:

Data is appended to the end of the table, rather than attempting to use existing free space within the table.

Data is written directly to the data files, by-passing the buffer cache.

*/

SQL> ANALYZE TABLE TEST_04 COMPUTE statistics;

```
SQL> ANALYZE TABLE TEST_04 COMPUTE statistics;

Table analyzed.
```

melakukan analisis dan perhitungan statistik lengkap (COMPUTE) pada tabel TEST_04

SQL> select blocks, pct_free , compression, compress_for

from   user_tables

where   table_name = 'TEST_04';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_04';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
        37          0 ENABLED  BASIC
```

mengambil dan menampilkan informasi tentang penggunaan blok penyimpanan (blocks), persentase ruang kosong (pct_free), serta status dan jenis kompresi (compression, compress_for) pada tabel TEST_04 yang diambil dari data dictionary view USER_TABLES

--case 5

--without compression, then change to compressed

SQL> create table test_05

as

select * from dba_objects where rownum <= 10000;

```
SQL> create table test_05
as
select * from dba_objects where rownum <= 10000;
  2    3
Table created.
```

membuat tabel baru bernama TEST_05 dengan menyalin struktur dan data dari data dictionary view DBA_OBJECTS, namun dibatasi hanya mengambil 10.000 baris data pertama

SQL> select blocks, pct_free , compression, compress_for

from   user_tables

where   table_name = 'TEST_05';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_05';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
                  10 DISABLED
```

mengambil dan menampilkan informasi tentang penggunaan blok penyimpanan (blocks), persentase ruang kosong (pct_free), serta status dan jenis kompresi (compression, compress_for) pada tabel TEST_05 yang diambil dari data dictionary view USER_TABLES

SQL> alter table test_05 compress basic;

```
SQL> alter table test_05 compress basic;


Table altered.
```

mengubah struktur tabel (ALTER TABLE) TEST_05 untuk mengaktifkan fitur Basic Compression

SQL> select blocks, pct_free , compression, compress_for

from   user_tables

where   table_name = 'TEST_05';

```
SQL> SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_05';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
                  10 ENABLED  BASIC
```

mengambil dan menampilkan informasi tentang penggunaan blok penyimpanan (blocks), persentase ruang kosong (pct_free), serta status dan jenis kompresi (compression, compress_for) pada tabel TEST_05 yang diambil dari data dictionary view USER_TABLES

SQL> alter table test_05 move;

```
SQL> alter table test_05 move;

Table altered.
```

melakukan perpindahan fisik (MOVE) data tabel TEST_05

SQL> ANALYZE TABLE TEST_05 COMPUTE statistics;

```
SQL> ANALYZE TABLE TEST_05 COMPUTE statistics;

Table analyzed.
```

melakukan analisis dan perhitungan statistik lengkap (COMPUTE) pada tabel TEST_05

SQL> select blocks, pct_free , compression, compress_for

from   user_tables

where   table_name = 'TEST_05';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_05';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
        37          0 ENABLED  BASIC
```

mengambil dan menampilkan informasi tentang penggunaan blok penyimpanan (blocks), persentase ruang kosong (pct_free), serta status dan jenis kompresi (compression, compress_for) pada tabel TEST_05 yang diambil dari data dictionary view USER_TABLES

==========

--Advanced row compression

/*

Rows inserted with or without using direct-path insert and updated rows

are compressed using advanced row compression.

Is enabled with CREATE TABLE … ROW STORE COMPRESS ADVANCED

Is recommended for active OLTP environments

With advanced row compression, duplicate values in the rows and columns in a data block are stored once

at the beginning of the block

note 1:

ROW STORE COMPRESS ADVANCED and COMPRESS BASIC are not supported for tables with more than 255

note 2: You cannot drop a column from a table that is compressed for direct-load operations,

although you can set such a column as unused.

*/

SQL> create table test_06

as

select * from dba_objects where rownum <= 20000;

```
SQL> create table test_06
as
select * from dba_objects where rownum <= 20000;
  2    3
Table created.
```

membuat tabel baru bernama TEST_06 dengan menyalin struktur dan data dari data dictionary view DBA_OBJECTS, namun dibatasi hanya mengambil 20.000 baris data pertama

SQL> ANALYZE TABLE test_06 COMPUTE statistics;

```
SQL> ANALYZE TABLE test_06 COMPUTE statistics;

Table analyzed.
```

melakukan analisis dan perhitungan statistik lengkap (COMPUTE) pada tabel TEST_06

SQL> select blocks, pct_free , compression, compress_for

from   user_tables

where   table_name = 'TEST_06';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_06';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
       365         10 DISABLED
```

mengambil dan menampilkan informasi tentang penggunaan blok penyimpanan (blocks), persentase ruang kosong (pct_free), serta status dan jenis kompresi (compression, compress_for) pada tabel TEST_06 yang diambil dari data dictionary view USER_TABLES

SQL> create table test_07 ROW STORE COMPRESS ADVANCED

as

select * from dba_objects where rownum <= 20000;

```
SQL> create table test_07 ROW STORE COMPRESS ADVANCED
as
select * from dba_objects where rownum <= 20000;
  2    3
Table created.
```

membuat tabel baru bernama TEST_07 dengan menyalin 20.000 baris data pertama dari view DBA_OBJECTS

SQL> ANALYZE TABLE test_07 COMPUTE statistics;

```
SQL> ANALYZE TABLE test_07 COMPUTE statistics;

Table analyzed.
```

melakukan analisis dan perhitungan statistik lengkap (COMPUTE) pada tabel TEST_07

SQL> select blocks, pct_free , compression, compress_for

from   user_tables

where   table_name = 'TEST_07';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_07';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
        87         10 ENABLED  ADVANCED
```

mengambil dan menampilkan informasi tentang penggunaan blok penyimpanan (blocks), persentase ruang kosong (pct_free), serta status dan jenis kompresi (compression, compress_for) pada tabel TEST_07 yang diambil dari data dictionary view USER_TABLES

--not let us try normal insert

SQL> create table test_08 ROW STORE COMPRESS ADVANCED

as

select * from dba_objects where rownum = 0;

```
SQL> create table test_08 ROW STORE COMPRESS ADVANCED
as
select * from dba_objects where rownum = 0;
  2    3
Table created.
```

membuat tabel baru bernama TEST_08 dengan mengaktifkan fitur ROW STORE COMPRESS ADVANCED (kompresi tingkat lanjut)

SQL> select * from test_08;

```
SQL> select * from test_08;

no rows selected
```

mengambil dan menampilkan semua data dari tabel TEST_08, no rows selected (kosong), karena

tabel ini dibuat tanpa menyalin data (hanya struktur kolom) menggunakan kondisi WHERE rownum = 0 sebelumnya

SQL> insert into test_08

select * from dba_objects where rownum <= 20000;

SQL> commit;

```
SQL> insert into test_08
select * from dba_objects where rownum <= 20000;
  2
20000 rows created.

SQL> commit;

Commit complete.
```

memasukkan (INSERT) 20.000 baris data dari view DBA_OBJECTS ke dalam tabel TEST_08

SQL> ANALYZE TABLE test_08 COMPUTE statistics;

```
SQL> ANALYZE TABLE test_08 COMPUTE statistics;

Table analyzed.
```

melakukan analisis dan perhitungan statistik lengkap (COMPUTE) pada tabel TEST_08

SQL> select blocks, pct_free , compression, compress_for

from   user_tables

where   table_name = 'TEST_08';

```
SQL> select  blocks, pct_free , compression, compress_for
from    user_tables
where   table_name = 'TEST_08';
  2    3
    BLOCKS   PCT_FREE COMPRESS COMPRESS_FOR
---------- ---------- -------- ------------------------------
        92         10 ENABLED  ADVANCED
```

mengambil dan menampilkan informasi tentang penggunaan blok penyimpanan (blocks), persentase ruang kosong (pct_free), serta status dan jenis kompresi (compression, compress_for) pada tabel TEST_08 yang diambil dari data dictionary view USER_TABLES

==========

SQL> show user;

SQL> show con_name;

```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

show user;: menampilkan user yang sedang terhubung

show con_name;: menampilkan sesi aktif di PDB/ CDB

SQL> CREATE TABLESPACE tbsalert

DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'

SIZE 50M LOGGING EXTENT MANAGEMENT LOCAL

SEGMENT SPACE MANAGEMENT AUTO;

```
SQL> CREATE TABLESPACE tbsalert
DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'
SIZE 50M  LOGGING EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;
  2    3    4
Tablespace created.
```

membuat tablespace baru bernama tbsalert yang menyimpan data fisiknya di file /opt/oracle/oradata/FREE/pdbts/tbsalert.dbf. Konfigurasi ini juga mengaktifkan LOGGING (pencatatan redo), EXTENT MANAGEMENT LOCAL (pengelolaan ekstensi lokal), dan SEGMENT SPACE MANAGEMENT AUTO (pengelolaan ruang segmen otomatis/ASSM)

SQL> select * from dba_data_files

where TABLESPACE_NAME='TBSALERT';

```
SQL> select * from dba_data_files
where TABLESPACE_NAME='TBSALERT';
  2
FILE_NAME
--------------------------------------------------------------------------------
   FILE_ID TABLESPACE_NAME                     BYTES     BLOCKS STATUS
---------- ------------------------------ ---------- ---------- ---------
RELATIVE_FNO AUT   MAXBYTES  MAXBLOCKS INCREMENT_BY USER_BYTES USER_BLOCKS
------------ --- ---------- ---------- ------------ ---------- -----------
ONLINE_ LOST_WR
------- -------
/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf
        20 TBSALERT                         52428800       6400 AVAILABLE
        1024 NO           0          0            0   46104576        5628
ONLINE  OFF
```

mengambil dan menampilkan semua informasi detail mengenai file data fisik (.dbf) yang terkait dengan tablespace TBSALERT yang baru dibuat, termasuk lokasi dan ukurannya. Informasi ini diambil dari data dictionary view DBA_DATA_FILES

SQL> select * from dba_free_space

where TABLESPACE_NAME='TBSALERT';

```
---------- ------------------------------ ---------- ---------- ---------
RELATIVE_FNO AUT   MAXBYTES  MAXBLOCKS INCREMENT_BY USER_BYTES USER_BLOCKS
------------ --- ---------- ---------- ------------ ---------- ------------
ONLINE_ LOST_WR
------- -------
/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf
        20 TBSALERT                         52428800       6400 AVAILABLE
        1024 NO           0          0            0   46104576        5628
ONLINE  OFF


SQL> select * from dba_free_space
where TABLESPACE_NAME='TBSALERT';

  2
TABLESPACE                     FILE_ID   BLOCK_ID      BYTES     BLOCKS
------------------------------ ---------- ---------- ---------- ----------
RELATIVE_FNO
------------
TBSALERT                              20        776   46071808       5624
        1024
```

mengambil dan menampilkan semua informasi tentang ruang kosong (free space) yang tersedia dalam bentuk ekstensi di dalam tablespace TBSALERT yang diambil dari data dictionary view DBA_FREE_SPACE

SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,

fs.bytes*100/ df.bytes PCT_FREE

FROM dba_data_files df ,dba_free_space fs

WHERE df.tablespace_name = fs.tablespace_name

AND df.tablespace_name = 'TBSALERT';

```
SQL> SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                       FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                       46071808   52428800     87.875
```

mengambil dan menampilkan informasi tentang total ruang tablespace, total ruang bebas, dan menghitung persentase ruang kosong (PCT_FREE) yang tersedia pada tablespace TBSALERT

SQL> begin

DBMS_SERVER_ALERT.SET_THRESHOLD(

metrics_id => dbms_server_alert.tablespace_pct_full,

warning_operator =>DBMS_SERVER_ALERT.OPERATOR_GE,

warning_value => '55',

critical_operator =>DBMS_SERVER_ALERT.OPERATOR_GE,

critical_value => '70',

observation_period => 1,

consecutive_occurrences => 1,

instance_name => 'free',

object_type =>DBMS_SERVER_ALERT.OBJECT_TYPE_TABLESPACE,

object_name => 'TBSALERT');

end;

```
SQL> begin
DBMS_SERVER_ALERT.SET_THRESHOLD(
metrics_id => dbms_server_alert.tablespace_pct_full,
warning_operator =>DBMS_SERVER_ALERT.OPERATOR_GE,
warning_value => '55',
critical_operator =>DBMS_SERVER_ALERT.OPERATOR_GE,
critical_value => '70',
observation_period => 1,
consecutive_occurrences => 1,
instance_name => 'free',
object_type =>DBMS_SERVER_ALERT.OBJECT_TYPE_TABLESPACE,
object_name => 'TBSALERT');
end;
  2    3    4    5    6    7    8    9   10   11   12   13   14  /

PL/SQL procedure successfully completed.
```

menjalankan blok kode PL/SQL untuk mengkonfigurasi ambang batas (threshold) peringatan server (server alert) otomatis pada tablespace TBSALERT

SQL> SELECT warning_value, critical_value

FROM dba_thresholds

WHERE object_name='TBSALERT';

```
SQL> SELECT warning_value, critical_value
FROM dba_thresholds
WHERE object_name='TBSALERT';
  2    3
WARNING_VALUE
--------------------------------------------------------------------------------
CRITICAL_VALUE
--------------------------------------------------------------------------------
55
70
```

mengambil dan menampilkan nilai ambang batas (threshold) Warning dan Critical yang telah dikonfigurasi untuk tablespace TBSALERT dari view DBA_THRESHOLDS

SQL> SELECT * FROM dba_outstanding_alerts

WHERE object_name='TBSALERT'; --

```
SQL> SELECT * FROM dba_outstanding_alerts
 WHERE object_name='TBSALERT';
  2
no rows selected
```

mengambil dan menampilkan semua informasi mengenai peringatan server (outstanding alerts) yang sedang aktif/berlangsung untuk tablespace TBSALERT yang diambil dari data dictionary view DBA_OUTSTANDING_ALERTS

SQL> create table test100 (emp_id number, name varchar2(100))

tablespace TBSALERT;

```
SQL> create table test100 ( emp_id number, name varchar2(100) )
 tablespace TBSALERT;
  2
Table created.
```

membuat tabel baru bernama TEST100 yang memiliki kolom emp_id dan name

SQL> begin

for i in 1..1000000

loop

insert into test100 values (i, 'any dummy name');

end loop;

commit;

end;

```
SQL> begin
 for i in 1..1000000
 loop
 insert into test100 values ( i, 'any dummy name' );
 end loop;
 commit;
 end;
  2    3    4    5    6    7    8  /

PL/SQL procedure successfully completed.
```

menjalankan blok kode PL/SQL untuk memasukkan (INSERT) data massal sebanyak 1.000.000 baris ke dalam tabel TEST100 yang berada di Tablespace TBSALERT menggunakan perulangan (loop)

SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,

fs.bytes*100/ df.bytes PCT_FREE

FROM dba_data_files df ,dba_free_space fs

WHERE df.tablespace_name = fs.tablespace_name

AND df.tablespace_name = 'TBSALERT';

```
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                       FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                         983040   52428800      1.875
TBSALERT                       15728640   52428800         30
```

mengambil dan menampilkan informasi mengenai total ruang tablespace, total ruang bebas, dan menghitung persentase ruang kosong (PCT_FREE) yang tersedia pada tablespace TBSALERT

--Wait a few minutes. 10-15

SQL> SELECT reason, message_type , message_level

FROM dba_outstanding_alerts

WHERE object_name='TBSALERT';

```
SQL> SELECT reason, message_type , message_level
FROM dba_outstanding_alerts
WHERE object_name='TBSALERT';

  2    3
REASON
--------------------------------------------------------------------------------
MESSAGE_TYPE MESSAGE_LEVEL
------------ -------------
Tablespace [TBSALERT@FREEPDB1] is [68 percent] full
Warning                 5
```

mengambil dan menampilkan rincian peringatan server yang sedang aktif/berlangsung untuk tablespace TBSALERT, termasuk alasan (reason), tipe pesan (message_type), dan level pesan (message_level) dari peringatan yang terpicu

==========

SQL> show user;

SQL> show con_name;

```
SQL> SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

show user;: menampilkan user yang sedang terhubung

show con_name;: menampilkan sesi aktif di PDB/ CDB

SQL> drop TABLESPACE tbsalert INCLUDING CONTENTS and DATAFILES;

```
SQL> drop TABLESPACE tbsalert INCLUDING CONTENTS and DATAFILES;

Tablespace dropped.
```

menghapus (DROP) tablespace bernama tbsalert secara permanen

SQL> CREATE TABLESPACE tbsalert

DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'

SIZE 50M LOGGING EXTENT MANAGEMENT LOCAL

SEGMENT SPACE MANAGEMENT AUTO;

```
SQL> CREATE TABLESPACE tbsalert
DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'
SIZE 50M  LOGGING EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;
  2    3    4
Tablespace created.
```

membuat tablespace baru bernama tbsalert yang menyimpan data fisiknya di file /opt/oracle/oradata/FREE/pdbts/tbsalert.dbf

SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,

fs.bytes*100/ df.bytes PCT_FREE

FROM dba_data_files df ,dba_free_space fs

WHERE df.tablespace_name = fs.tablespace_name

AND df.tablespace_name = 'TBSALERT';

```
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                     FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                     46071808   52428800     87.875
```

mengambil dan menampilkan informasi mengenai total ruang tablespace, total ruang bebas, dan menghitung persentase ruang kosong (PCT_FREE) yang tersedia pada tablespace TBSALERT

SQL> create table test100 (emp_id number, name varchar2(100))

tablespace TBSALERT;

```
SQL> create table test100 ( emp_id number, name varchar2(100) )
 tablespace TBSALERT;
  2
Table created.
```

membuat tabel baru bernama TEST100 yang memiliki kolom emp_id dan name

SQL> begin

for i in 1..1000000

loop

insert into test100 values (i, 'any dummy name');

end loop;

commit;

end;

```
SQL> begin
 for i in 1..1000000
 loop
 insert into test100 values ( i, 'any dummy name' );
 end loop;
 commit;
 end;
  2    3    4    5    6    7    8  /

PL/SQL procedure successfully completed.
```

menjalankan blok kode PL/SQL untuk memasukkan (INSERT) data sebanyak 1.000.000 baris ke dalam tabel TEST100 yang berada di Tablespace TBSALERT menggunakan perulangan (loop)

SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,

fs.bytes*100/ df.bytes PCT_FREE

FROM dba_data_files df ,dba_free_space fs

WHERE df.tablespace_name = fs.tablespace_name

AND df.tablespace_name = 'TBSALERT';

```
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                       FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                         983040   52428800      1.875
TBSALERT                       15728640   52428800         30
```

mengambil dan menampilkan informasi mengenai total ruang tablespace, total ruang bebas, dan menghitung persentase ruang kosong (PCT_FREE) yang tersedia pada tablespace TBSALERT

SQL> delete from test100;

SQL> commit;

```
SQL> delete from test100;

1000000 rows deleted.

SQL> commit;

Commit complete.
```

menghapus semua baris data (1.000.000 baris) dari tabel TEST100, kemudian lakukan commit untuk menyimpan dan mengakhiri transaksi, memastikan perubahan menjadi permanen

SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,

fs.bytes*100/ df.bytes PCT_FREE

FROM dba_data_files df ,dba_free_space fs

WHERE df.tablespace_name = fs.tablespace_name

AND df.tablespace_name = 'TBSALERT';

```
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                       FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                         983040   52428800      1.875
TBSALERT                       15728640   52428800         30
```

mengambil dan menampilkan informasi tentang total ruang tablespace, total ruang bebas, dan menghitung persentase ruang kosong (PCT_FREE) yang tersedia pada tablespace TBSALERT

--Create a Segment Advisor Task

SQL> DECLARE

tname VARCHAR2(128) := 'my_seg_task3';

tname_desc VARCHAR2(128) := 'Get shrink advice for segments in TBSALERT';

task_id NUMBER;

object_id NUMBER;

objectname VARCHAR2(100);

objecttype VARCHAR2(100);

BEGIN

dbms_advisor.create_task('Segment Advisor', task_id,tname,tname_desc,NULL);

dbms_advisor.create_object(tname,'TABLESPACE','TBSALERT',' ',' ',NULL, ' ',object_id) ;

dbms_advisor.set_task_parameter(tname,'RECOMMEND_ALL','TRUE');

END;

```
SQL> DECLARE
 tname VARCHAR2(128) := 'my_seg_task3';
 tname_desc VARCHAR2(128) := 'Get shrink advice for segments in TBSALERT';
 task_id NUMBER;
 object_id NUMBER;
 objectname VARCHAR2(100);
 objecttype VARCHAR2(100);
 BEGIN
 dbms_advisor.create_task('Segment Advisor', task_id,tname,tname_desc,NULL);
 dbms_advisor.create_object(tname,'TABLESPACE','TBSALERT',' ',' ',NULL, ' ',obje
ct_id) ;
 dbms_advisor.set_task_parameter(tname,'RECOMMEND_ALL','TRUE');
 END;
  2    3    4    5    6    7    8    9   10   11   12   13  /

PL/SQL procedure successfully completed.
```

menjalankan blok kode PL/SQL untuk membuat dan mengkonfigurasi tugas (*task*) baru bernama my_seg_task3 pada Segment Advisor

--Execute the task.

SQL> DECLARE

tname VARCHAR2(128) := 'my_seg_task3';

BEGIN

dbms_advisor.EXECUTE_TASK(tname);

END;

```
SQL> DECLARE
tname VARCHAR2(128) := 'my_seg_task3';
BEGIN
dbms_advisor.EXECUTE_TASK(tname);
END;
  2    3    4    5    6  /

PL/SQL procedure successfully completed.
```

menjalankan blok kode PL/SQL untuk mengeksekusi tugas (EXECUTE_TASK) bernama my_seg_task3 pada Segment Advisor

SQL> SELECT DESCRIPTION FROM dba_advisor_tasks WHERE TASK_NAME='my_seg_task3';

```
SQL> SELECT DESCRIPTION FROM dba_advisor_tasks WHERE TASK_NAME='my_seg_task3';

DESCRIPTION
--------------------------------------------------------------------------------
Get shrink advice for segments in TBSALERT
```

mengambil dan menampilkan kolom DESCRIPTION (deskripsi/tujuan tugas) dari view DBA_ADVISOR_TASKS untuk memverifikasi detail dan tujuan dari tugas Segment Advisor bernama my_seg_task3 yang telah dikonfigurasi sebelumnya

SQL> SELECT attr1, attr2, message

FROM dba_advisor_findings f, dba_advisor_objects o

WHERE f.task_name = o.task_name

AND f.object_id = o.object_id

AND f.task_name = 'my_seg_task3';

```
SQL> SELECT attr1, attr2, message
FROM dba_advisor_findings f, dba_advisor_objects o
 WHERE f.task_name = o.task_name
 AND f.object_id = o.object_id
 AND f.task_name = 'my_seg_task3';
  2    3    4    5
ATTR1
--------------------------------------------------------------------------------
ATTR2
--------------------------------------------------------------------------------
MESSAGE
--------------------------------------------------------------------------------
SYS
TEST100
Enable row movement of the table SYS.TEST100 and perform shrink, estimated savin
gs is 29351936 bytes.
```

mengambil dan menampilkan hasil temuan (*findings*) dan rekomendasi (*message*) dari tugas Segment Advisor bernama my_seg_task3

SQL> alter table test100 enable row movement;

```
SQL> alter table test100 enable row movement;

Table altered.
```

mengubah konfigurasi tabel TEST100 untuk mengaktifkan fitur ROW MOVEMENT

SQL> ALTER TABLE test100 SHRINK SPACE;

```
SQL> ALTER TABLE test100 SHRINK SPACE;

Table altered.
```

mengubah tabel TEST100 dengan menjalankan perintah SHRINK SPACE (penyusutan segmen)

SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,

fs.bytes*100/ df.bytes PCT_FREE

FROM dba_data_files df ,dba_free_space fs

WHERE df.tablespace_name = fs.tablespace_name

AND df.tablespace_name = 'TBSALERT';

```
SQL> SELECT df.tablespace_name tablespace, fs.bytes free, df.bytes,
fs.bytes*100/ df.bytes PCT_FREE
FROM dba_data_files df ,dba_free_space fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = 'TBSALERT';
  2    3    4    5
TABLESPACE                       FREE      BYTES   PCT_FREE
------------------------------ ---------- ---------- ----------
TBSALERT                       46006272   52428800      87.75
```

mengambil dan menampilkan informasi tentang total ruang tablespace, total ruang bebas, dan menghitung persentase ruang kosong (PCT_FREE) yang tersedia pada tablespace TBSALERT

==========

/*

Managing Resumable Space Allocation

A resumable statement:

1- Enables you to suspend large operations instead of receiving an error

2- Gives you a chance to fix the problem while the operation is suspended, rather

than starting over

3-Is suspended for the following conditions:

>>Out of space

>>Maximum extents reached

>>Space quota exceeded

4-Can be suspended and resumed multiple times

*/

SQL> show user;

SQL> show con_name;

```
SQL> show user;
USER is "SYS"
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
```

show user;: Menampilkan user yang sedang terhubung

show con_name;: Menampilkan user saat ini terhubung ke PDB/ CDB

SQL> drop TABLESPACE tbsalert INCLUDING CONTENTS and DATAFILES;

```
SQL> drop TABLESPACE tbsalert INCLUDING CONTENTS and DATAFILES;

Tablespace dropped.
```

menghapus (DROP) tablespace bernama tbsalert secara permanen. INCLUDING CONTENTS memastikan seluruh objek (tabel, index, dll) di dalamnya ikut terhapus, dan opsi AND DATAFILES menghapus file fisik (.dbf) yang terkait dari sistem operasi

SQL> CREATE TABLESPACE tbsalert

DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'

SIZE 10M LOGGING EXTENT MANAGEMENT LOCAL

SEGMENT SPACE MANAGEMENT AUTO;

```
SQL> CREATE TABLESPACE tbsalert
DATAFILE '/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'
SIZE 10M  LOGGING EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;
  2    3    4
Tablespace created.
```

membuat tablespace baru bernama tbsalert yang menyimpan data fisiknya di file /opt/oracle/oradata/FREE/pdbts/tbsalert.dbf dengan ukuran awal 10MB. Konfigurasi ini juga mengaktifkan LOGGING (pencatatan redo), EXTENT MANAGEMENT LOCAL (pengelolaan ekstensi secara lokal), dan SEGMENT SPACE MANAGEMENT AUTO

SQL> create table test100 (emp_id number, name varchar2(100))

tablespace TBSALERT;

```
SQL> create table test100 ( emp_id number, name varchar2(100) )
 tablespace TBSALERT;
  2
Table created.
```

membuat tabel baru bernama TEST100 yang memiliki kolom emp_id dan name

SQL> begin

for i in 1..400000

loop

insert into test100 values (i, 'any dummy name');

commit;

end loop;

end;

```
SQL> begin
 for i in 1..400000
 loop
 insert into test100 values ( i, 'any dummy name' );
  commit;
 end loop;
end;
  2    3    4    5    6    7    8  /
begin
*
ERROR at line 1:
ORA-01653: unable to increase tablespace TBSALERT by 1MB during insert or
update on table SYS.TEST100
ORA-06512: at line 4
Help: https://docs.oracle.com/error-help/db/ora-01653/
```

menjalankan blok kode PL/SQL untuk memasukkan (INSERT) data sebanyak 400.000 baris ke dalam tabel TEST100, error terjadi karena Tablespace TBSALERT sudah penuh

SQL> select * from test100;

```
----------
NAME
--------------------------------------------------------------------------------
     87398
any dummy name

     87399
any dummy name

     87400
any dummy name


    EMP_ID
----------
NAME
--------------------------------------------------------------------------------
     87401
any dummy name


107701 rows selected.
```

mengambil dan menampilkan semua data dari tabel TEST100

/*

A statement executes in resumable mode only if its session has been enabled by

one of the following actions:

1- The RESUMABLE_TIMEOUT initialization parameter is set to a nonzero value.

2- An ALTER SESSION ENABLE RESUMABLE statement is issued

*/

SQL> ALTER SESSION ENABLE RESUMABLE;

```
SQL> ALTER SESSION ENABLE RESUMABLE;

Session altered.
```

mengubah (ALTER) pengaturan sesi saat ini untuk mengaktifkan fitur RESUMABLE (Resumable Space Allocation)

SQL> truncate table test100;

```
SQL> truncate table test100;

Table truncated.
```

mengosongkan (TRUNCATE) tabel TEST100 dengan menghapus semua baris data secara permanen

SQL> begin

for i in 1..400000

loop

insert into test100 values (i, 'any dummy name');

commit;

end loop;

end;

```
SQL> begin
 for i in 1..400000
 loop
 insert into test100 values ( i, 'any dummy name' );
  2    3    4    5
  6  commit;
  7  end loop;
  8  end;
  9  /

PL/SQL procedure successfully completed.
```

--the script will be suspend 7,200 seconds=2 hours. (this the default)

--you can do ALTER SESSION ENABLE RESUMABLE TIMEOUT 3600

--now oracle will not give us any message that this script has been suspend

--but we can see it as DBA. open new session and do this

/*

SQL> select * from dba_resumable;

```
SQL> select * from dba_resumable;


PL/SQL procedure successfully completed.
```

mengambil dan menampilkan semua informasi lengkap tentang sesi atau operasi DML/DDL yang sedang ditangguhkan (suspended) karena mengalami masalah kekurangan ruang penyimpanan (tablespace full)

SQL> SELECT file_name, autoextensible FROM dba_data_files WHERE

tablespace_name='TBSALERT';

```
SQL> SELECT file_name, autoextensible FROM dba_data_files WHERE
tablespace_name='TBSALERT';

PL/SQL procedure successfully completed.
```

mengambil dan menampilkan nama file fisik (file_name) dan status AUTOEXTENSIBLE (apakah file data dapat bertambah ukurannya secara otomatis ketika penuh) dari tablespace TBSALERT, yang diambil dari data dictionary view DBA_DATA_FILES

SQL> ALTER DATABASE DATAFILE

'/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf'

AUTOEXTEND ON MAXSIZE unlimited;

```
--------------------------------------------------------------------------------
ERROR_PARAMETER3
--------------------------------------------------------------------------------
ERROR_PARAMETER4
--------------------------------------------------------------------------------
ERROR_PARAMETER5
--------------------------------------------------------------------------------
ERROR_MSG
--------------------------------------------------------------------------------


SQL> SQL>   2
FILE_NAME
--------------------------------------------------------------------------------
AUT
---
/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf
YES


SQL>   2    3
Database altered.
```

mengubah (ALTER) konfigurasi file data fisik /opt/oracle/oradata/FREE/pdbts/tbsalert.dbf (milik tablespace TBSALERT)

SQL> SELECT file_name, autoextensible FROM dba_data_files WHERE

tablespace_name='TBSALERT';

```
SQL> SELECT file_name, autoextensible FROM dba_data_files WHERE
tablespace_name='TBSALERT';
  2
FILE_NAME
--------------------------------------------------------------------------------
AUT
---
/opt/oracle/oradata/FREE/pdbts/tbsalert.dbf
YES
```

mengambil dan menampilkan nama file fisik (file_name) dan status AUTOEXTENSIBLE (apakah file data dapat bertambah ukurannya secara otomatis ketika penuh) dari tablespace TBSALERT, yang diambil dari data dictionary view DBA_DATA_FILES

--WAIT until NO ROWS retrived by THIS QUERY

SQL> select * from dba_resumable;

```
---------- --------- ---------- ---------- ---------- ---------
   TIMEOUT START_TIME           SUSPEND_TIME         RESUME_TIME
---------- ------------------- ------------------- -------------------
NAME
--------------------------------------------------------------------------------
SQL_TEXT
--------------------------------------------------------------------------------
ERROR_NUMBER
------------
ERROR_PARAMETER1
--------------------------------------------------------------------------------
ERROR_PARAMETER2
--------------------------------------------------------------------------------
ERROR_PARAMETER3
--------------------------------------------------------------------------------
ERROR_PARAMETER4
--------------------------------------------------------------------------------
ERROR_PARAMETER5
--------------------------------------------------------------------------------
ERROR_MSG
--------------------------------------------------------------------------------
```

mengambil dan menampilkan semua informasi lengkap tentang sesi atau operasi DML/DDL yang sedang ditangguhkan (suspended) karena mengalami masalah kekurangan ruang penyimpanan (tablespace full), diambil dari view DBA_RESUMABLE dan hanya akan menunjukkan data jika fitur RESUMABLE diaktifkan pada sesi yang menjalankan operasi tersebut

SQL> select count(1) from test100;

```
SQL> select count(1) from test100;

  COUNT(1)
----------
   2800000
```

mengambil dan menampilkan jumlah total baris data yang ada di dalam tabel TEST100

*/